

Open Science & FAIR

Introduction to Data Management Practices course

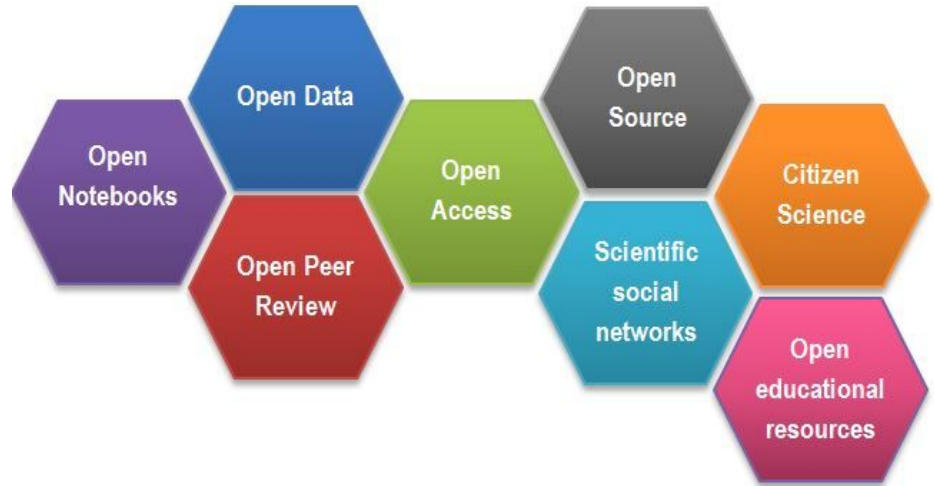
NBIS DM Team

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Open Science

Make scientific research and its dissemination accessible to all levels of society.

- Open methodology
- Open source
- **Open data**
- Open access
- Open peer review
- Open educational resources



"Open Science facets as a beehive" by Gema
Bueno de la Fuente licenced under [CC-BY](#)

UNESCO Open Science Recommendation & Toolkit

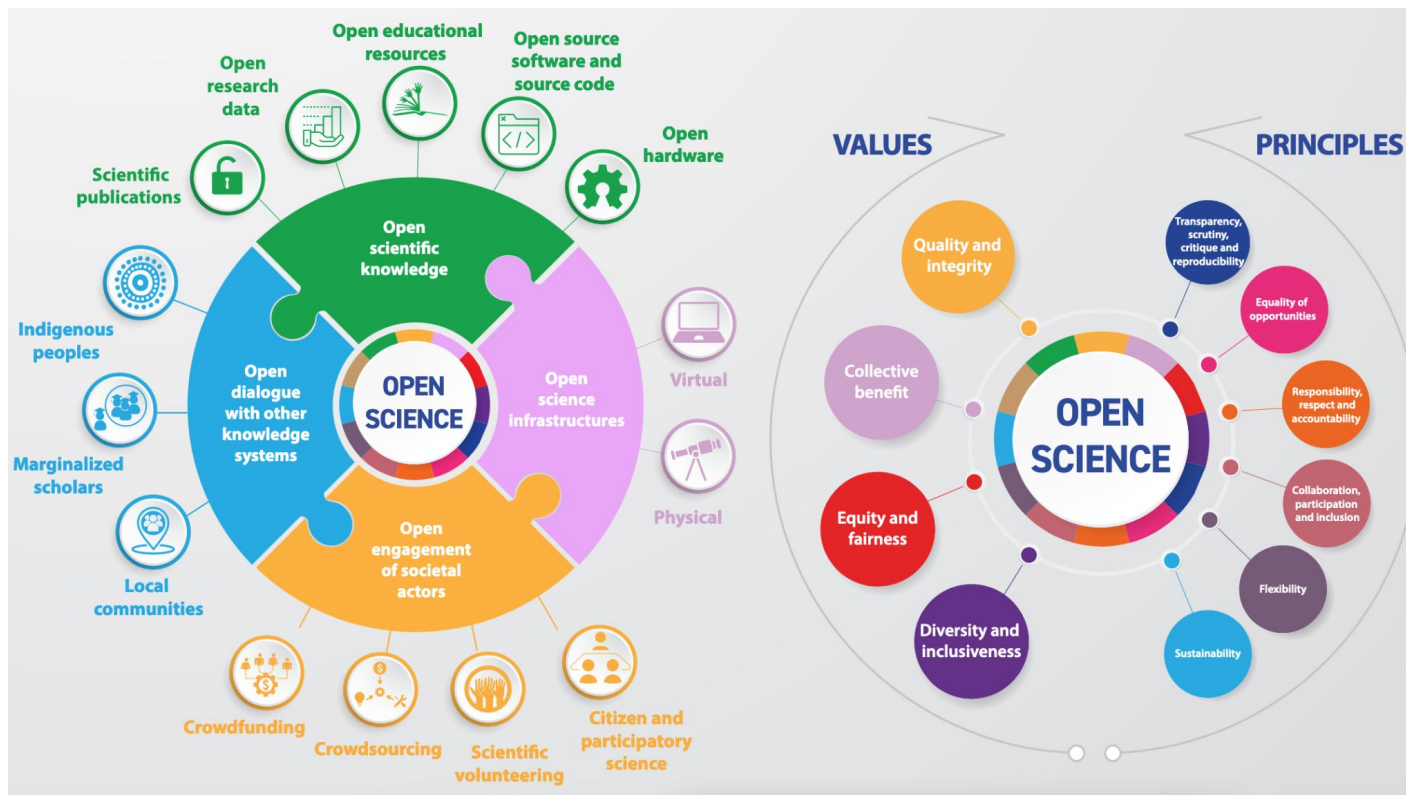


Illustration from [UNESCO open science toolkit: contents](#) under a [Creative Commons Attribution Share-Alike 3.0 License](#)

**What do you think are
reasons for Open Data?**

Open Data

- Democracy and transparency
 - Publicly funded research data should be accessible to all
 - Published results and conclusions should be possible to check by others
- Research
 - Enables others to combine data, address new questions, and develop new analytical methods
 - Reduce duplication and waste
- Innovation and utilization outside research
 - Public authorities, companies, and private persons outside research can make use of the data
- Citation
 - Citation of data will be a merit for the researcher that produced it



Ethical?

Doing “sloppy” science & not being open and transparent

Waste of resources

Contributing to the current research **credibility** crisis

Contributing to the current **reproducibility** crisis

Harming the profession

Harming public trust in research

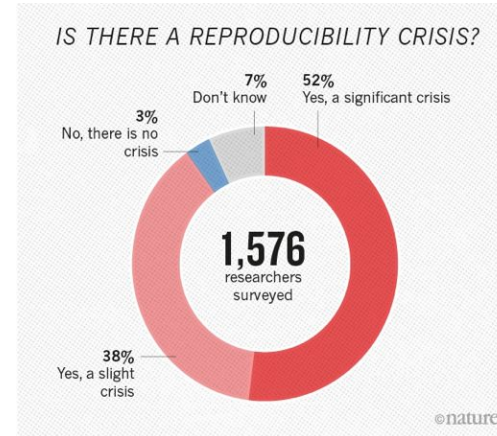
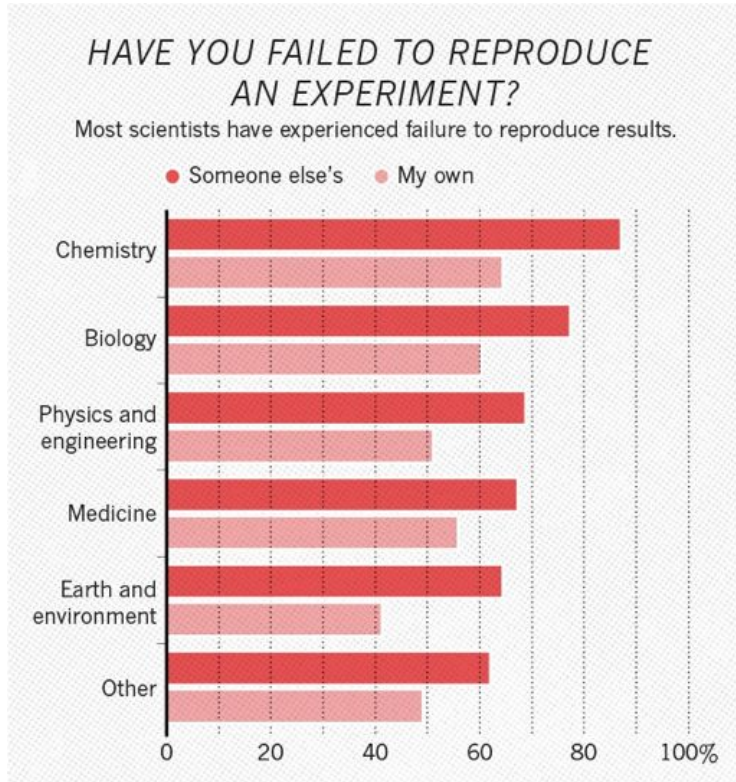


Crisis?

Do you think we have a **reproducibility** crisis?

If so, what are some of its causes?

A reproducibility crisis



[1] "1,500 scientists lift the lid on reproducibility". *Nature*. 533: 452–454

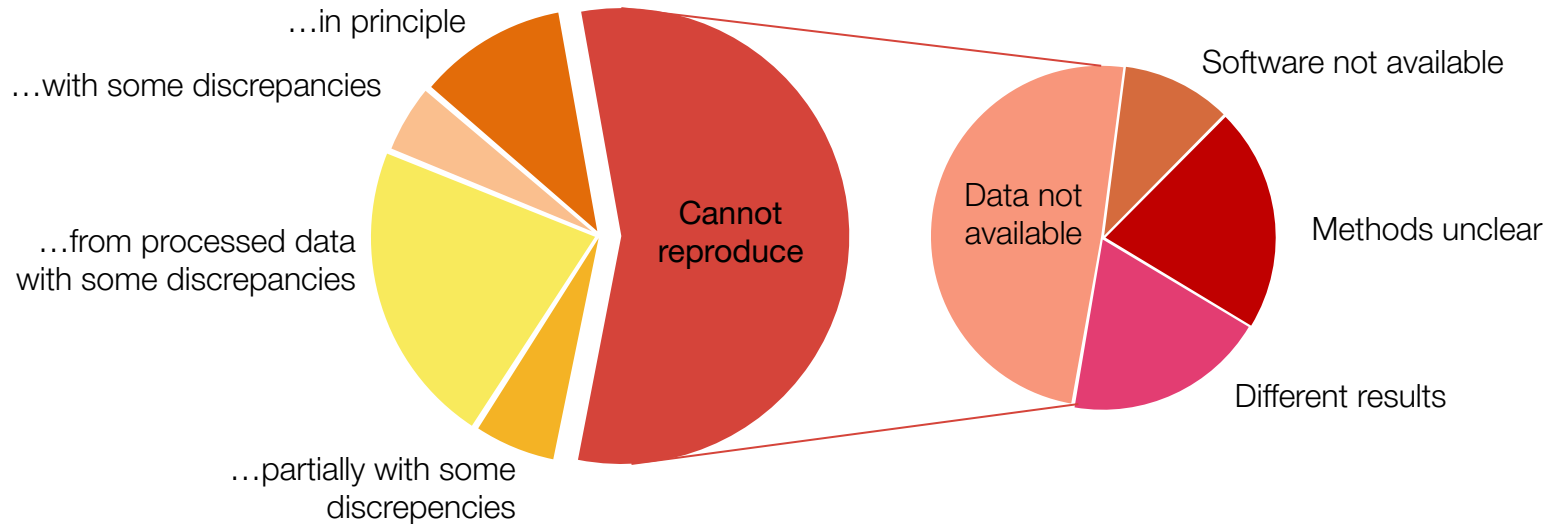
[2] Begley, C. G.; Ellis, L. M. (2012). "Drug development: Raise standards for preclinical cancer research". *Nature*. 483 (7391): 531–533.



A reproducibility crisis

Reproduction of data analyses in 18 articles on microarray-based gene expression profiling published in Nature Genetics in 2005–2006:

Can reproduce...



Summary of the efforts to replicate the published analyses.

Adopted from: Ioannidis et al. Repeatability of published microarray gene expression analyses. *Nature Genetics* 41 (2009) doi:10.1038/ng.295

Data Management Snafu



<https://www.youtube.com/watch?v=N2zK3sAtr-4>

FAIR

- To be useful for others data should be:
 - **FAIR** - Findable, Accessible, Interoperable, and Reusable
... for both Machines and Humans

Wilkinson, Mark et al. “The FAIR Guiding Principles for scientific data management and stewardship”. Scientific Data 3, Article number: 160018 (2016) <http://dx.doi.org/10.1038/sdata.2016.18>



DOI: 10.1038/sdata.2016.18

Box 2 | The FAIR Guiding Principles

To be Findable:

- F1. (meta)data are assigned a globally unique and persistent identifier
- F2. data are described with rich metadata (defined by R1 below)
- F3. metadata clearly and explicitly include the identifier of the data it describes
- F4. (meta)data are registered or indexed in a searchable resource

To be Accessible:

- A1. (meta)data are retrievable by their identifier using a standardized communications protocol
- A1.1 the protocol is open, free, and universally implementable
- A1.2 the protocol allows for an authentication and authorization procedure, where necessary
- A2. metadata are accessible, even when the data are no longer available

To be Interoperable:

- I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- I2. (meta)data use vocabularies that follow FAIR principles
- I3. (meta)data include qualified references to other (meta)data

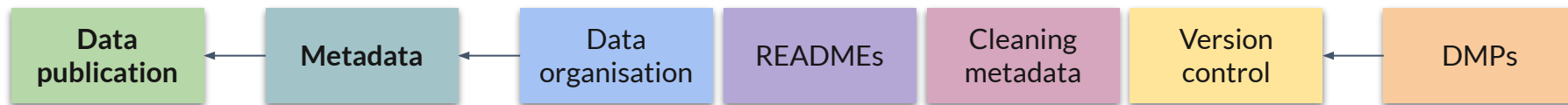
To be Reusable:

- R1. (meta)data are richly described with a plurality of accurate and relevant attributes
- R1.1. (meta)data are released with a clear and accessible data usage license
- R1.2. (meta)data are associated with detailed provenance
- R1.3. (meta)data meet domain-relevant community standards



Findable

- Data have a **globally unique persistent identifier**
 - *e.g. a DOI, database accession number, etc*
- Data are described by **metadata**
 - *Information that explains the data*
- Data and metadata are findable in a **search resource**
 - *There must be ways of searching for the data*



Metadata?

The data about the data (or anything really)

Example: Spleen samples from mice

organism	strain	age	time unit	developmental stage	sex	organism part	individual	genotype
Mus musculus	C57BL/6J x 129Sv	12	week	adult	male	spleen	287 - 2	HellsF/F Mb1Cre+
Mus musculus	C57BL/6J x 129Sv	12	week	adult	male	spleen	287 - 2	HellsF/F Mb1Cre+
Mus musculus	C57BL/6J x 129Sv	15	week	adult	female	spleen	278 - 6	HellsF/F Mb1Cre+
Mus musculus	C57BL/6J x 129Sv	15	week	adult	female	spleen	278 - 6	HellsF/F Mb1Cre+

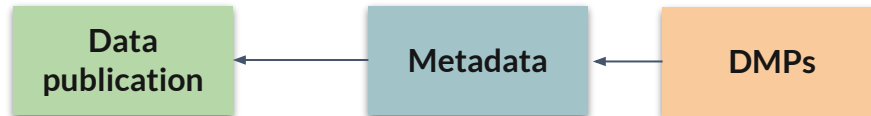
Accessible

- Data is retrievable through a **standardised communication protocol** (open, free, allowing authentication & authorisation where necessary)
 - *e.g. http, sftp, etc*
- Metadata are accessible, **even if data is no longer available**
 - *Information about the data can be found even if data is no longer available*



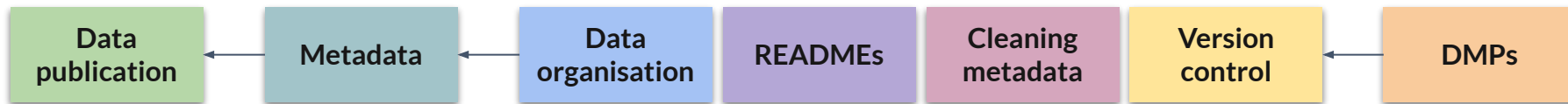
Interoperable

- Metadata use a formal, accessible, shared **language for knowledge representation**
 - *Metadata is available in a form that even a computer can make use of*
- Metadata use **vocabularies** that follow the FAIR principles
 - *Standardised ways of capturing information about the data (that are in themselves FAIR)*
- Metadata include qualified **references** to other metadata
 - *If the data relies on other data, there must be links to those*



Reusable

- Data have a clear **data usage license**
 - *It is obvious under what conditions the data can be reused*
- Metadata are associated with **detailed provenance**
 - *The metadata is detailed enough to understand for what research questions it is relevant to reuse*
- Metadata meet domain-relevant community **standards**
 - *Metadata is described according to existing standards in the research field*



FAIR

- Both humans and machines are intended users of data
- The principles are not necessarily about *open* data
 - “As open as possible, as closed as necessary”
- FAIRness is not something absolute
 - Different levels of FAIR maturity
- FAIR does not enforce any particular technical standards

Simple FAIR data example



"mus musculus" AND "rna-seq of coding rna" AND "time series d
Examples: E-MEXP-31, cancer

ArrayExpress HomeBrowseSubmitHelpAbout BioStudiesFeedback

ArrayExpressFunctional genomics data

BIOSTUDIES / ARRAYEXPRESS

Study type
see all

☐ animal - single-cell sequencing

☐ cell line - high-throughput sequencing

☐ clip-seq

☐ rna-seq of coding rna

☐ rna-seq of coding rna from single cells

Experimental Design
TOP 10
see all

☐ case control design

☐ co-expression_design

☐ development or differentiation design

☐ disease state design

☐ genetic modification design

☐ genotype design

☐ in_vivo_design

☐ stimulus or stress design

☐ time series design

☐ time_series_design

Organism
see all

☐ mouse

Search results for "mus musculus" AND "rna-seq of coding rna" AND "time series design" AND liver

1 – 10 of 10 results

Sort by: Relevance

E-MTAB-7020 • 13 November 2018 • 3 links • 2 files • 76 views

[RNA-seq study on time of day specific Glucocorticoid action in mouse liver and lung tissues](#)

...E-MTAB-7020 RNA-seq study on time of day specific Glucocorticoid action in mouse liver and lung tissues RNA-seq of coding RNA EFO EFO_0003738 **Mus musculus** Glucocorticoids are critical regulators of energy metabolism and immunity and major drug targets in inflammatory disease. It is also known...

E-MTAB-10239 • 19 May 2021 • 1 link • 2 files • 120 views

[scRNA-seq of murine mucosal associated invariant T \(MAIT\) cells after Francisella tularensis infection](#)

...E-MTAB-10239 scRNA-seq of murine mucosal associated invariant T (MAIT) cells after Francisella tularensis infection Animal - Single-cell sequencing RNA-seq of coding RNA from single cells EFO EFO_0005684 **Mus musculus** Mucosal-Associated Invariant T (MAIT) cells are an innate-like subset of alpha...

E-MTAB-7017 • 13 November 2018 • 3 links • 2 files • 122 views

[RNASeq data analysis of wild type and reverb alpha knockout cells from mouse liver, at different time points, with or without DEX treatment](#)

...E-MTAB-7017 RNASeq data analysis of wild type and reverb alpha knockout cells from mouse liver, at different time points, with or without DEX treatment RNA-seq of coding RNA EFO EFO_0003738 **Mus musculus** Synthetic glucocorticoids are the most potent anti-inflammatory agents known and are widely used...

E-MTAB-1538 • 8 March 2013 • 1 link • 2 files • 40 views

[mouse APAP](#)

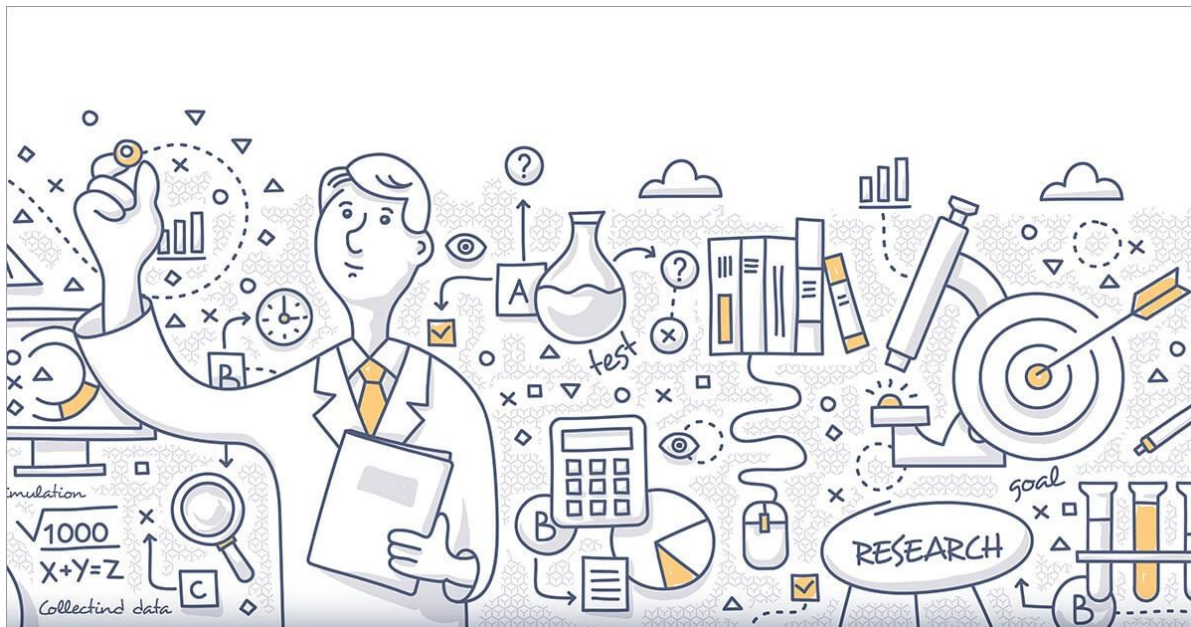
...E-MTAB-1538 mouse_APAP RNA-seq of coding RNA EFO EFO_0003738 **Mus musculus** Comprehensive cataloging of cell-free circulating RNAs using next generation sequencing technology may open a window to assess drug associated adverse effects at the systems level. To explore this potential, we conducted...

Picture source: [ArrayExpress @ EMBL-EBI](#)



When to be FAIR?

FAIR at source?



© rassco

Retroactively?

Good Data Management Practices

- **Data Management Plans**, to do your thinking ahead of time
- **Using standard metadata descriptions**, to clearly define your data
- **Organising your outputs**, so you and others can understand what you have done
- **Use versioning control** to keep track of changes you do
- **Clean up metadata and data** to be consistent with the standards you have chosen
- **Submit your data to international public repositories**, so others can find and reuse your data

What data management practices do you apply in your research projects today?

	Ad Hoc	One-Time	Active and Informative	Optimized for Re-Use
Planning your project	When it comes to my data, I have a "way of doing things" but no standard or documented plans.	I create some formal plans about how I will manage my data, but I generally don't refer back to them.	I develop detailed plans about how I will manage my data that I actively revisit and revise over the course of a project.	I design my plans for managing data to streamline future use by myself or others.
Organizing your data	I don't follow a consistent approach for keeping my data organized, so it often takes time to find things.	I have an approach for organizing my data, but I only put it into action after my project is complete.	I have an approach for organizing my data that I implement prospectively, but it not necessarily standardized.	I organize my data to the so that others can navigate, understand, and use it without me being present.
Saving and backing up your data	I decide what data is important while I am working on it and typically save it in a single location.	I know what data needs to be saved and I back it up after I'm done working on it to reduce the risk of loss.	I have a system for regularly saving important data while I am working on it. I have multiple backups.	I save my data in a manner and location designed maximize opportunities for re-use by myself and others.
Getting your data ready for analysis	I don't have a standardized or well documented process for preparing my data for analysis.	I have thought about how I will need to prepare my data, but I handle each case in a different manner.	My process for preparing data is standardized and well documented.	I prepare my data in such a way as to facilitate use by both myself and others in the future.
Analyzing your data and handling the outputs	I often have to redo my analyses or examine their products to determine what procedures or parameters were applied.	After I finish my analysis, I document the specific parameters, procedures, and protocols applied.	I regularly report the specifics of both my analysis workflow and decision making process while I am analyzing my data.	I have ensured that the specifics of my analysis workflow and decision making process can be put into action by others.
Sharing and publishing your data	I share the results of my research, but generally I do not share the underlying data.	I share my my data only when I'm required to do so or in response to direct requests from other researchers.	I regularly share the data that underlies my results and conclusions in a form that enables use by others.	Because of my excellent data management practices, I am able to efficiently share my data whenever I need to with whomever I need to.

Borghi, J. et al (2018). *Support your Data*.
<https://doi.org/10.3897/rio.4.e26439>

Voting

	Ad Hoc	One-Time	Active and Informative	Optimized for Re-Use
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https://bit.ly/support_your_data_rubric

The Political landscape



The Political Landscape

- Policymakers are **pushing for research data to be made available** as openly as possible
- Big investments are being made in **infrastructure and skills for data sharing and reuse**



Policy landscape

- [UNESCO's Recommendation on Open Science](#)
- EU
 - Open Science Policy, Directives
 - European Research Council, Horizon Europe
 - EOSC
- Swedish research policy bills 2016, 2020, 2024
 - Transition to open research data by 2026 & push for FAIR
 - Government assignments to KB & VR
- [SUHF national roadmap for open science](#)
- [University policies](#)
- [National guidelines for open science](#) (KB)

Open Science

FAIR

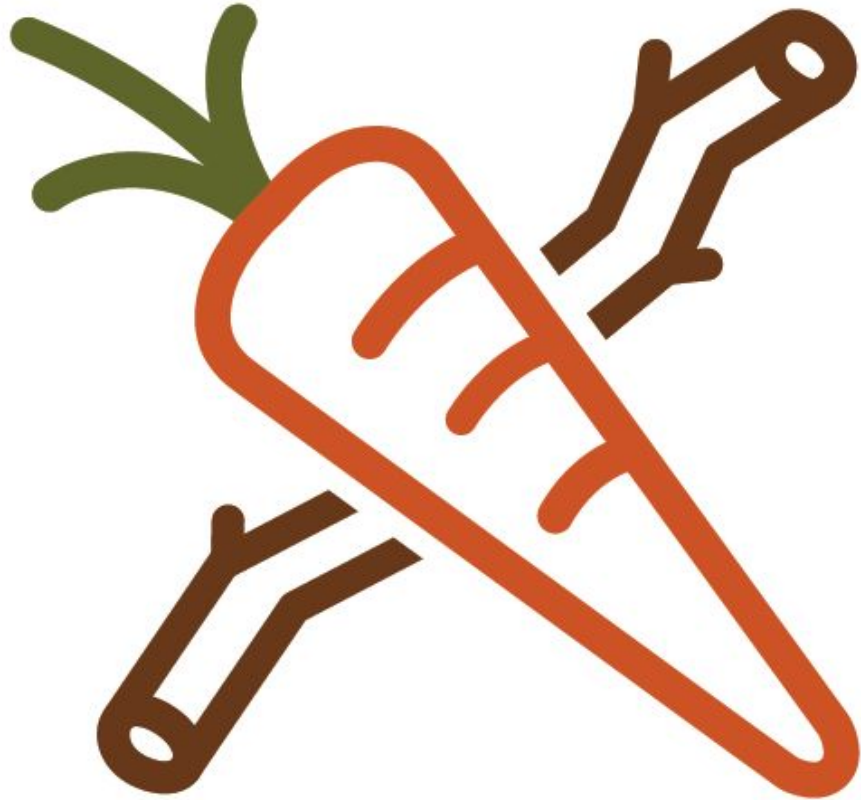
*“As open as possible,
as closed as necessary”*

*“FAIR [...] open data sharing should
become the default [...]”*

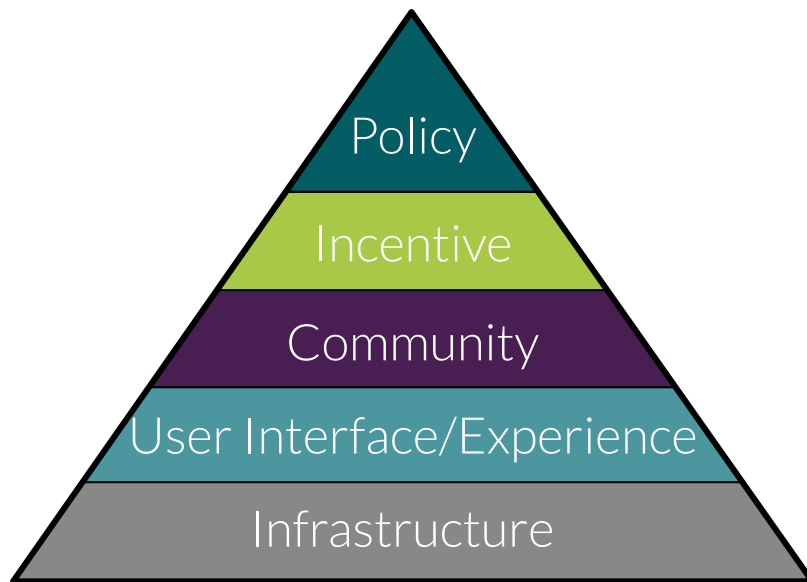
*“[...] research results [...] meet the
FAIR principles and can be reused
by everyone”*



Motivators



Open Science: from policy to practice



Make it required

Make it rewarding

Make it normative

Make it easy

Make it possible

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